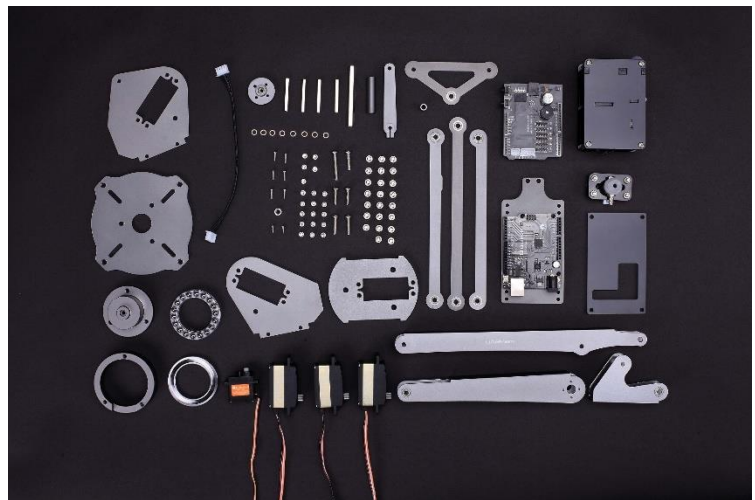




# uArm Metal Developer Guide

---

2017-03



**UFACTORY**

UFACTORY TECHNOLOGY CO., LTD | NANSHAN DISTRICT, SHENZHEN, P. R. CHINA  
WWW.UFACTORY.CC

# Contents

|                                                     |    |
|-----------------------------------------------------|----|
| • Introduction .....                                | 2  |
| • Software.....                                     | 3  |
| • Software Installation .....                       | 3  |
| • Communication Protocol .....                      | 5  |
| • Calibration .....                                 | 5  |
| • Hardware .....                                    | 6  |
| uArm Metal Body .....                               | 6  |
| • Basic Information of uArm Metal Body.....         | 6  |
| • uArm metal Specifications .....                   | 10 |
| • Board-Arduino.....                                | 11 |
| • Board Reference.....                              | 11 |
| • Pinout.....                                       | 12 |
| • Control Board Diagram .....                       | 12 |
| • Board Specification .....                         | 12 |
| • Servo .....                                       | 14 |
| • Basic Information of Servo.....                   | 14 |
| • Servo Specification .....                         | 16 |
| • The definition of port for uArm Metal servo ..... | 17 |
| • Control .....                                     | 18 |
| • Vacuum Pump .....                                 | 19 |
| • Reference .....                                   | 20 |
| • Protection.....                                   | 20 |



## Introduction

Maybe now you are a  
Software Engineer  
Mechanism Engineer  
Electronic Engineer ;

Maybe now you are studying the  
Artificial Intelligence  
the Computer Science the  
Machine Learning ;

Maybe now you are A  
Scientist  
An Artist ;

Anyway, if you are reading this Guide-

You are a developer who has interest in ROBOTICS!

We are here to help you build your creative, fun, smart robot ideas come true with uArm! MAKERS COME  
FROM ALL JACKS OF LIFE, SO DO UARM MASTERS!

# Software

## Software Installation

UFACTORY provide a link for you that you can download all documents about uArm Metal:

<https://drive.google.com/open?id=0B-L-tCvknXU9bUpPbEp5Ry1SYXM>

### 1. Basic Software Installation

Please refer to [Getting Started Guide](#).

### 2. Developer Software Installation

#### Windows

As Python is not defaulted inside of Windows, if you want to use these developing tools, please **DOWNLOAD and INSTALL Python** first.

Next, please install **pip**. We recommend [pip-for-windows](#).

Now you can install uArm' tool kit by sending the command - `pip install pyuarm`.

#### Mac

What you need:

`Pip -avrdude`

Strongly recommend to use this, because it is a really very convenient tool that you type only one install command:

This command will help you with the installation and very efficient:)

```
bash -c "$(curl -fsSLhttp://download.ufactory.cc/tools/macosx/install.sh)"
```

If you would like to install by yourself, you can do so:

First, install pip - `sudo easy_install pip`

Next, install pyuarm - `pip install pyuarm`

If you need to upgrade the firmware, you need to install `avrdude`

-refer to Homebrew, a quick and easy way to make it.

Completing homebrew, you can start `brew install avrdude`.

#### Linux

What you need:

`pip -avrdude`

You can use the *installation tool* in the released version of Linux, for example, in the *debain*, you can just use:

```
sudo apt-get install python-pip python-dev build-essentialavrdude
```

After installation, you can use: - `pip install pyuarm` installation tools.

## Source Code and API Documentation

arduino IDE: <https://www.arduino.cc/en/Main/Software>

**PLEASE DO NOT download your Arduino IDE from <https://www.arduino.org> , because you may not be able to use that. We recommend Arduino IDE V1.6.12 or later.**

arduino API: <http://uarm-developer.github.io/UArmForArduino/index.html>

Python API: <http://pyuarm.readthedocs.io/en/latest/>

Pyuarm API: <https://pypi.python.org/pypi/pyuarm>

uArm protocol: <https://github.com/uArm-Developer/UArmForArduino/blob/master/PROTOCOL.md>

Arduino UNO board intro: <https://www.arduino.cc/en/Main/ArduinoBoardUno?setlang=cn>

uArm Metal 3D document: <https://grabcad.com/library/uarm-metal-1>

## Communication Protocol

### uArm Metal's Communication Protocol

We write uArm's Communication Protocol in order to help you program with your preferred languages at PC.

Once you send the command (via uArm Protocol), it will automatically analyze it and act as related. If you would like to write your own uArm protocol for communication, please refer to our [communication Protocol on guithub](#).

## Calibration

### Why calibration?

The analog data ADC adopted is not accurate, what's more, the inaccuracy of analog data is different from each other, so we need to conduct the liner calibration once.

### How to calibrate by yourself?

It was calibrated before sale.

1. Please DON'T calibrate it if not necessary. All uArms were calibrated by us before sale. Frequent calibration will cause damage to the uArm.
2. If you have to calibrate, Please use **uarmAssistant** , function **Re-calibration**.
3. uarmAssistant is limited to the purchase of uArm users before April 2016, **after April 2016, the user does not have calibration problems, DO NOT DO THIS STEP.**

## Hardware

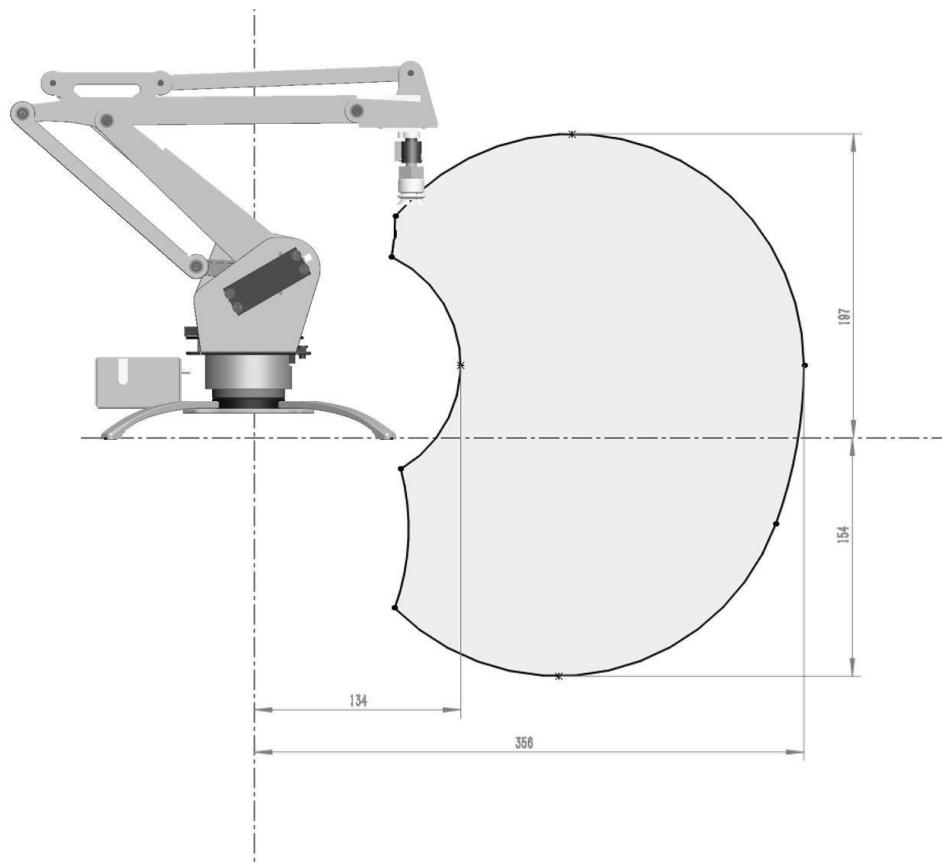
### uArm Metal Body

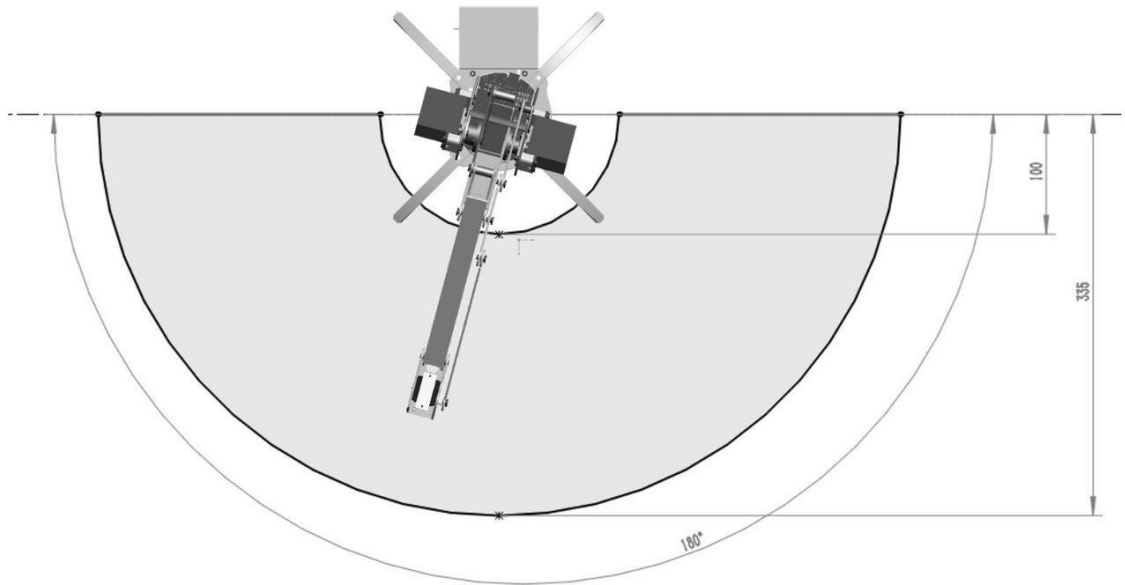
#### Basic Information of uArm Metal Body

##### Work Range

You might have guessed the work range with the Limits of XYZ, and the work range for uArm is not a regular area. However, you might not know there are somewhere which is out of the work range, so we have set restrains for those places which uArm cannot reach, and return an Error. *WorkRangeDiagramofuArm*

##### Work Range PIC 1





### **Work Range PIC2**

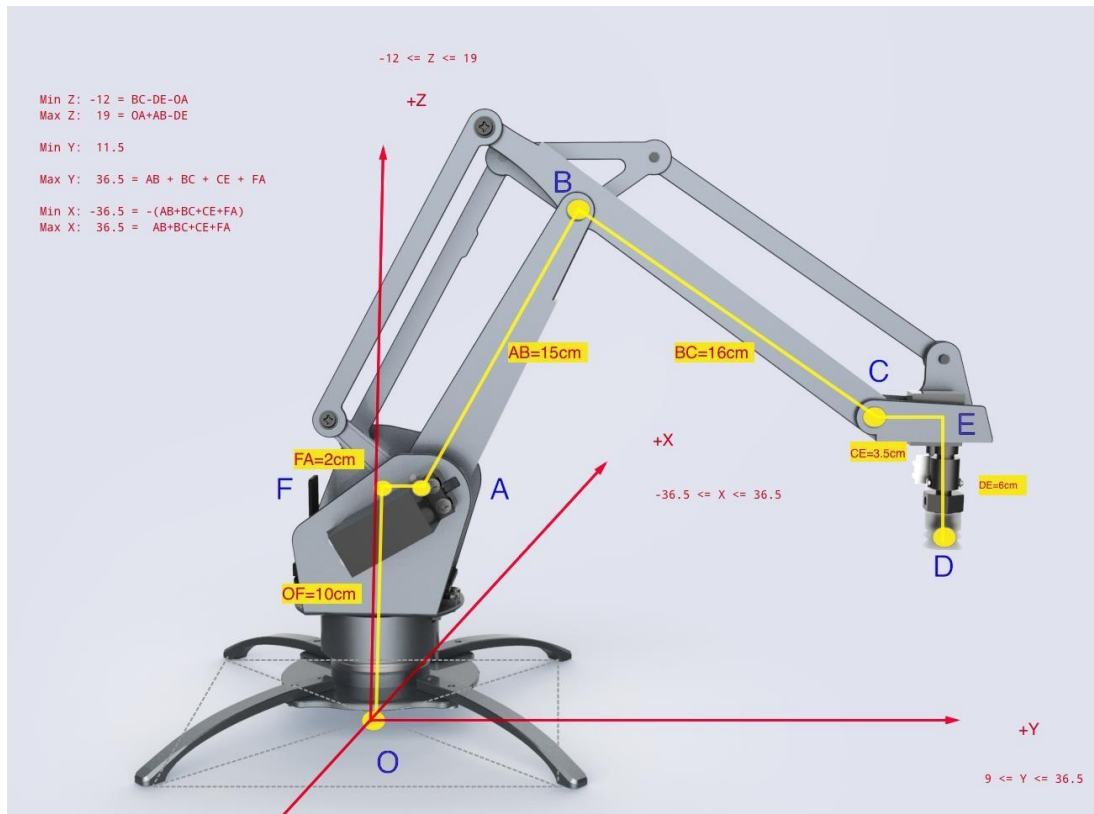
### **Accuracy and Offset**

Theoretically, the offset is around  $\pm 0.5\text{CM}$ . In fact, the range is related with the payload for uArm. Servo has less offset when working in the range between  $40^\circ$  and  $140^\circ$ ; but when it is less than  $40^\circ$  or greater than  $140^\circ$ , the offset will become wider.

So, please control the payload, don't make uArm work out of its range, and pay attention not to push servo working a wide range beyond its limit.

### **Three Dimension coordination**

We use mathematical model- Three Dimension Coordination to describe uArm's work range. Please refer to the pic below Dimension Diagram



**PIC Three Dimension Diagram**

### 1) Point of Junction

**O** Original Point (0,0,0)

**A** The middle point of Left/ Right Axis

**B** The middle point between two screws--each screw is correspondently on one side of Link 2 as the linkage of Point B.

**C** The middle point between two screws-- each screw is correspondently on one side of Link 2 as the linkage of Point C.

**D** The Center Point on the bottom of Suction Cup.

**E** It is an interaction of DE and CE, which are perpendicular to each other. Like in the picture, DE is paralleled with coordinate axis Z and CE is paralleled with coordinate axis Y.

**F** It is an interaction of OF and AF, which are perpendicular to each other. Like in the picture, OF

is paralleled with coordinate axis Z and AF is paralleled with coordinate axis Y.

## **2) Line**

**OA** Distance of Original Point O to the center of L/R axis A

**OF** = 10CM Shadow of OA on coordinate axis +Z **AF** = 2CM

Shadow of OA on coordinate axis +Y **AB** = 14.8 CM Length  
of Link2

**BC** = 16CM Length of Link1

**CE** = 3.5 CM Length of Front Part

**DE** = 6CM Length of Suction Cup

**Extreme/Limits** = [-36.5cm ~ 36.5cm]

**Max X** = AB + BC + CD + AF [Y] =

[11.5 cm ~ 36.5cm] **MIN Y** = Min Y

**Max Y** = Max X

[Z] = [-12cm ~ 19cm] **Max**

**Z** = OA + AB -DE **Min Z** = BC

- DE – OA

## uArm metal Specifications

### 1. Apply Environmental Condition

| No. | Item                      | Specification |
|-----|---------------------------|---------------|
| 1.1 | Storage Temperature Range | -20°C ~ 70°C  |
| 1.2 | Operate Temperature Range | 0°C ~ 30°C    |

### 2. Standard Test Environment

| No. | Item              | Specification |
|-----|-------------------|---------------|
| 2.1 | Temperature Range | -1 0°C ~ 60°C |
| 2.2 | Humidity Range    | 65%±1 0%      |

### 3. Mechanical Specification

| No. | Item               | Specification    |
|-----|--------------------|------------------|
| 3.1 | Size               | 300*270*110mm    |
| 3.2 | Weight             | 1.9kg            |
| 3.3 | Material           | Aluminum         |
| 3.4 | Water Resistance   | NO               |
| 3.5 | Repeatability      | 10mm(Maximum)    |
| 3.6 | Action Radius      | 12CM~32CM        |
| 3.7 | Max Lifting Weight | 500g (with 15kg) |
| 3.8 | Accuracy           | 6~10MM (typical) |
| 3.9 | Lifetime           | >100000 times    |

### 4. Electrical Specification

| No. | Item            | Specification |
|-----|-----------------|---------------|
| 4.1 | Operat voltage  | DC5V          |
| 4.2 | Idle current    | 200MA         |
| 4.3 | working current | 3.5A          |

## Board-Arduino

### Board Reference

We use Arduino UNO to modify but have not change the basic working theory. You can learn from the [Arduino Official Webcote](#), if you are interested.

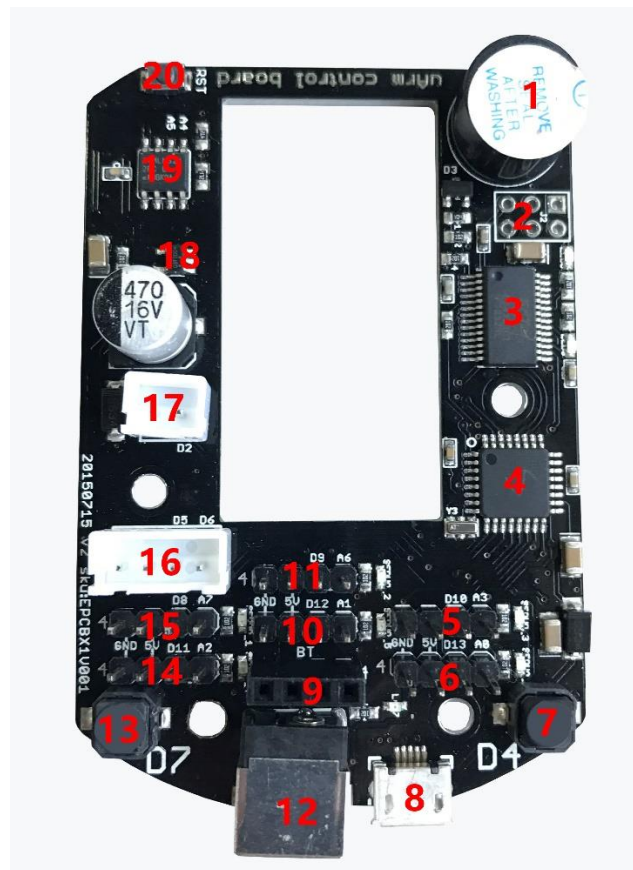
### Board Pinout

[Pic link](#) of pinout (You can refer to “Developer Pack”/Arduino/ PIC of “pinout”).

### Control Board Diagram

As for Control Board, please [Check the Link here](#). (You can refer to “Developer Pack”/Arduino/ PIC of “uArm-Control Board.”)

### Board Specification



| No. | Item                        | Description                                                                                                                                                           |
|-----|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | buzzer                      | Sound reminder- to remind you the position of uArm, port D3-PD3(INT1).                                                                                                |
| 2   | ISP download port           | Starting in the upper left corner in clockwise order: RESET-PC6(RESET)->D13-PB5(SCK)->D12-PB4(MISO)-> VCC(+5V)-> D11-PB3(MOSI)->GND; PB/PC Port is in the ATmega328p. |
| 3   | USB converts to Serial port | Online debug; communication-FT232RL                                                                                                                                   |
| 4   | ATmega328p                  | - Master chip                                                                                                                                                         |

|    |                                                         |                                                                                                                            |
|----|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| 5  | Servo 3 Connector<br>(Control the arm end of the servo) | From Left to Right:<br>GND->VCC(+5V)->D10- PB2(SS)->A3-PC3(ADC3);<br>D10 is PWM output and A3 is analog input.             |
| 6  | Servo 1 Connector<br>(Control the left servo)           | From Left to Right:<br>GND->VCC(+5V)->D13-PB5(SCK)->A0-PC0(ADC0);<br>D10 is PWM output and A3 is analog input.             |
| 7  | Switch D4-PD4(T0)                                       | uArm condition exchange.                                                                                                   |
| 8  | MINIUSB-A                                               | Download software and in-out communication.                                                                                |
| 9  | Bluetooth port                                          | From Left to Right:<br>D1-PD1(TXD)->D0-PD0(RXD)->VCC(+5V)->GND;<br>uArm condition and specification setting and operation. |
| 10 | Servo 0 Connector<br>(Control the base servo)           | From Left to Right:<br>GND->VCC(+5V)->D12-PB4(MISO)->A1PC1(ADC1);<br>D12 is PWM output and A1 is analog input.             |
| 11 | Servo Connector<br>(Control the electric griper)        | From Left to Right:<br>GND->VCC(+5V)->D9-PB1(OC1)->A6-(ADC6);<br>D9 is PWM output and A6 is analog input.                  |
| 12 | Power                                                   | 5V power adapter and input currency >5A                                                                                    |
| 13 | Switch D7-PD7(AIN1)                                     | Change uArm condition.                                                                                                     |
| 14 | Servo 2 Connector<br>(Control the right servo)          | From Left to Right:<br>GND->VCC(+5V)->D11- PB3(MOSI)->A2-PC2(ADC2);<br>D11 is PWM output and A2 is analog input.           |
| 15 | Servo connector ( preserved )                           | From Left to Right:<br>GND->VCC(+5V)->D8- PB0(ICP)->A7-(ADC7);<br>D8 is PWM output and A7 is analog input.                 |
| 16 | Pump connector                                          | From Left to Right:<br>GND->VCC(+5V)->D8- PB0(ICP)->A7-(ADC7). D8 is PWM output and A7 is analog input.                    |
| 17 | End suction tip limit switch interface                  | From Left to Right:<br>GND->D2                                                                                             |
| 18 | LDO                                                     | 5V transfer 3.3V                                                                                                           |
| 19 | EEPROM External power-down memory                       | 64KB                                                                                                                       |
| 20 | Reset                                                   |                                                                                                                            |

## Servo

### Basic Information of Servo

#### Servo Schematics

When we talk about the servo control, we have to mention Arduino. Thanks to Arduino, we can have thousand open sourced libraries. For uArm, our servo motor supports

Arduino's *servo.h* library.

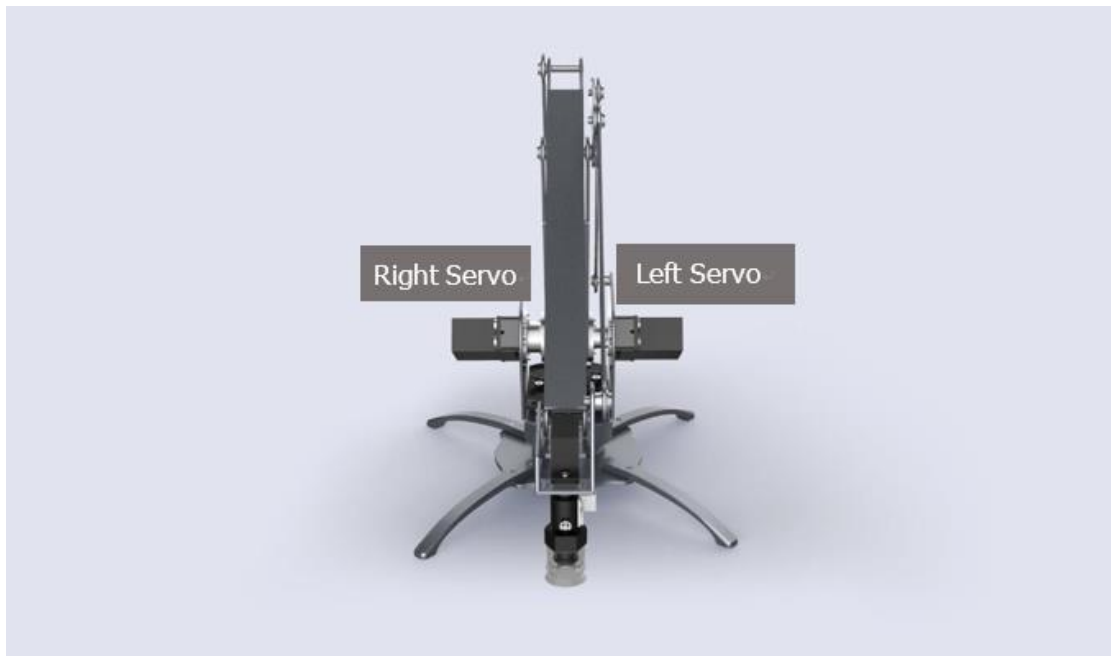
#### Servo Attach/Detach

Before using servo, it is a must to attach servos. At this time, servos will be locked.

#### Rotation Ranges of Servo

We use default functions for servo *servoWrite()*. The unit of its rotation is *degree*, ranging from 0 to 180.

When assembling, we install the servo to uArm with a fixed angle.



The servo on the left can range from 0° to 150° and the one on the right is 20° to 150°.

Now you can control the movement of uArm by read the servo's angle, but this is not visible, and somehow dangerous. Because the linkage of servos may lock each other, it is a factor lead to burn the servo. We would like to suggest you to construct a three dimension coordination to control uArm' move.

## Read the servo angle

There is a build-in ADC (analog and digital convertor), which can change the current analog signal into digital one. And then, with the function analogRead which is defaulted in Arduino, the digital data can be read out and get the current angle with this

formula  $intercept + analog * slope$ .

## Servo Specification

### 1. Condition

|                             |              |
|-----------------------------|--------------|
| Storage Temperature Range   | -20°C ~ 70°C |
| Operating Temperature Range | -10°C ~ 60°C |

### 2. Mechanism

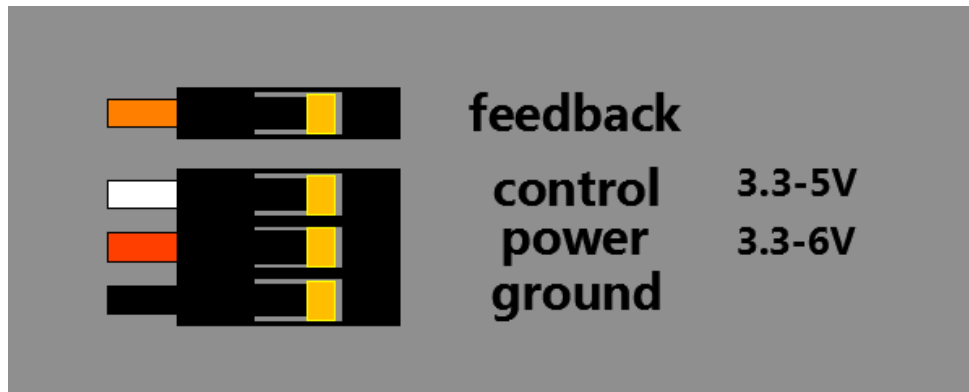
|                         |                                 |
|-------------------------|---------------------------------|
| Size                    | 40×20×38mm                      |
| Weight                  | 56g                             |
| Gear type               | metal gear                      |
| Limit angle             | 210°±5°                         |
| Bearing                 | 2BB                             |
| Horn gear spline        | 25T                             |
| Horn type               | Plastic, POM                    |
| Case                    | Engineering plastics(Polyamide) |
| Motor                   | DC motor                        |
| Splash water resistance | No                              |

### 3. Electric

|                         |                                 |
|-------------------------|---------------------------------|
| Horn type               | Plastic, POM                    |
| Case                    | Engineering plastics(Polyamide) |
| Connector wire          | 300mm±5mm                       |
| Motor                   | DC motor                        |
| Splash water resistance | No                              |

## The definition of port for uArm Metal servo

DM1500A is fully compactable with normal 3 pin port and the forth is used for the output of potentiometer as position feedback of uArm.



## Voltage and angle

| An  | Value (TYP) |
|-----|-------------|
| 0°  | 0.364V      |
| 90  | 1.723V      |
| 180 | 3.076V      |

Ps:1, because of the correspondence-difference of potentiometer inside servo, different servo might have  $\pm 10\%$  difference.

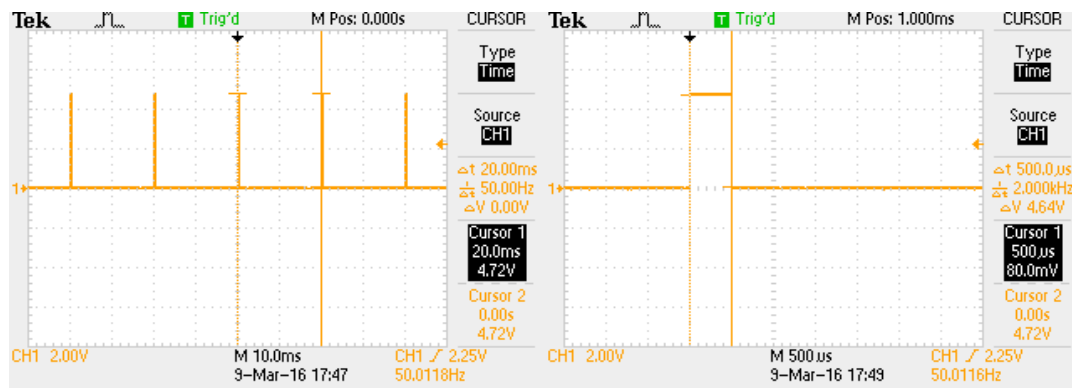
2, Different ADC will leads different sample number when output.

## Control

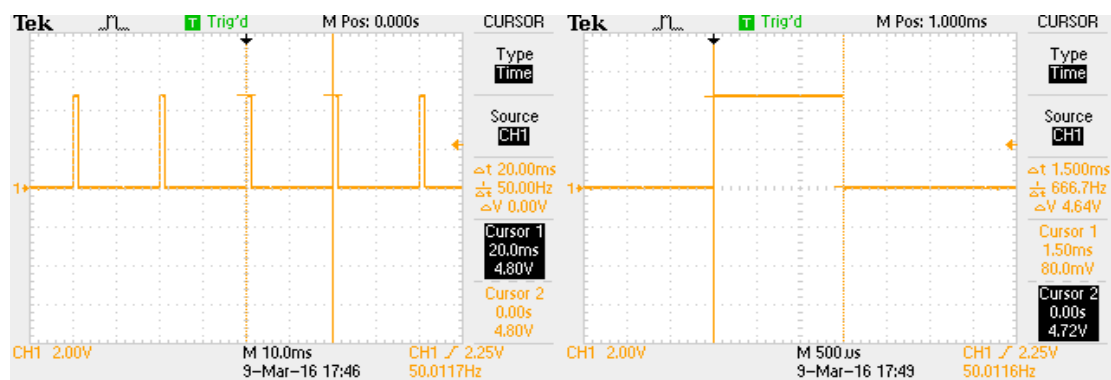
PWM and specification as following:

|                           |                                                   |
|---------------------------|---------------------------------------------------|
| <b>Command signal</b>     | Pulse width modification                          |
| <b>Pulse frequency</b>    | 50Hz                                              |
| <b>Pulse width range</b>  | 500 ~ 2500usec                                    |
| <b>Neutral position</b>   | 1500usec                                          |
| <b>Running degree</b>     | $180^{\circ} \pm 3^{\circ}$ (when 500 ~ 2500usec) |
| <b>Dead band width</b>    | 2 usec                                            |
| <b>Rotating direction</b> | Clock wise(when 500 ~ 2500usec)                   |

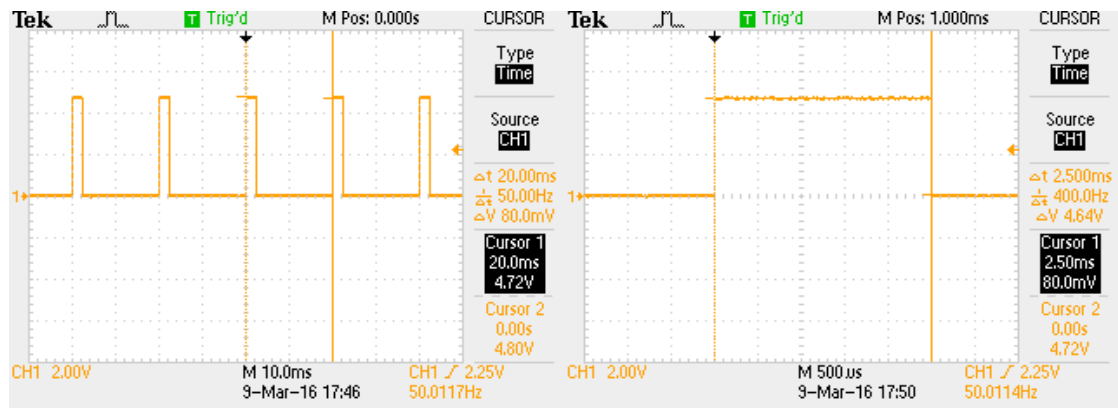
When control port is  $0^{\circ}$ , the oscillo-gram is the pic below (the pic on the right is enlarged drawing of a single wave).



When control port is  $90^{\circ}$ , the oscillo-gram is the pic below (the pic on the right is enlarged drawing of a single wave)



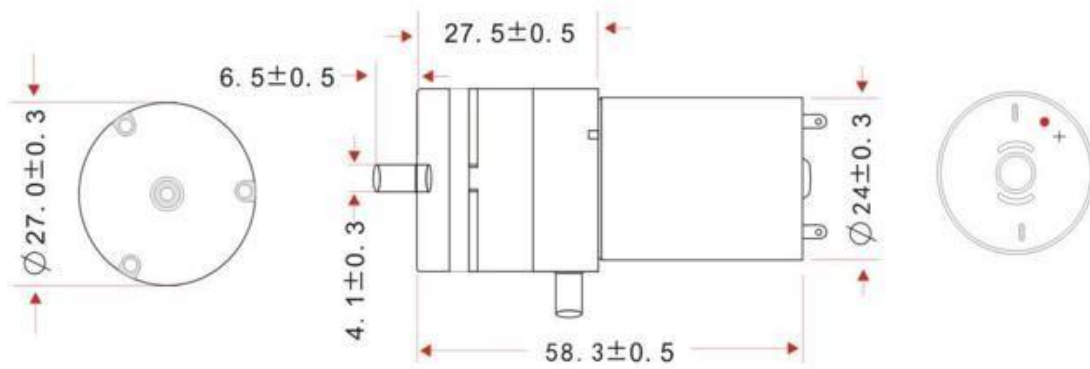
When control port is  $180^\circ$ , the oscillo-gram is the pic below (the pic on the right is enlarged drawing of a single wave).





## Vacuum Pump/mechanical Gripper/ Universal Holder

### Pump



**Voltage :** DC 4-5V

**Current :** <400mA

### Reference

1. [Pump Specification](#) (You can refer to “Developer Pack”/Pump/ “Pump Specification”.)
2. [Pump User Guide](#) (You can refer to “Developer Pack”/Pump/ “uArm Vacuum System User Guide”.)

[Gripper](#) (You can refer to “Developer Pack”/Gripper/ “gripper user guide”.)

[Universal Holder](#) (You can refer to “Developer Pack”/Gripper/ “holder user guide”.)



## Protection

### How to prevent damaging uArm?

Make sure you use our firmware and it is updated, the payload and work range is not go beyond of its limits.

### Under what circumstances will the servo be burned or damaged?

1, The payload is out of its limit or work with the load almost its limit for a long time; 2,

The servo is hit hard;

3, Use the wrong power supply, which often burn the servo.

### Under what circumstances should I unplug uArm ASAP?

When you see the red light is flashing, which means something wrong already happen, please unplug ASAP.



## 5. uArm Community

Welcome to the uArm Community! Now, you can explore uArm with the world of robot enthusiasts, Makers, artists and scientists!

### Official Website:

<https://www.ufactory.cc/>

### Official Forum:

<https://forum.ufactory.cc/>

### Official WeChat Account:



### Official Media:



More Help?

[【Contact Us】](#).